

The Blue Lotus Stress-Testing Engine: Design, Methodology, and Out-of-Sample Verification

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Abstract

We present the Blue Lotus Stress-Testing Engine, a system that maps a single return series to a forward-looking distribution of drawdown, tail-loss, and recovery outcomes. The engine integrates a deterministic three-state volatility regime model, Bayesian shrinkage of the regime parameters, Generalized Pareto tail estimation via Peaks-over-Threshold Extreme Value Theory, and regime-switching Monte Carlo path generation with a drawdown-conditional blend and a hard loss floor. An optional Esscher importance resampler and bootstrapped confidence intervals on every reported metric complete the pipeline. A Model Fragility Index, defined as the mean Wasserstein-1 distance between the baseline and parameter-perturbed drawdown distributions, provides a single auditable measure of the estimates' sensitivity to parameter uncertainty.

The method is evaluated under a strict out-of-sample protocol. For nine named market crises and ten liquid asset proxies, the engine is re-estimated using only the history available prior to each crisis and its predicted drawdown distribution is compared with the realized outcome, yielding 89 crisis cells; an identical protocol applied to 213 calm windows constitutes the calibration control. On calm data the 5% and 1% tail bounds are breached in 2.8% and 0.9% of cells respectively, neither rejectable by a Kupiec test ($p = 0.11$, $p = 0.93$). Under the nine crises these rates rise to 56% and 31%, degrading monotonically with severity and failing entirely only at the two fastest systemic shocks on record. The engine thus reproduces ordinary risk faithfully and reports, rather than conceals, the residual tail gap that history alone cannot close.

Keywords: stress testing, Monte Carlo, regime switching, Extreme Value Theory, Expected Shortfall, Esscher transform, Wasserstein distance, out-of-sample backtest, Kupiec test.

JEL: C15, C53, C58, G17, G32.

Disclaimer: the engine produces probability distributions, not point predictions or investment advice. All outputs are conditional on the modelling assumptions described herein.

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1. Introduction

1.1 Scope and objective

Given a history of daily returns — a market index, a fund, or the profit-and-loss of a trading strategy — the engine addresses a single question: conditional on the return-generating behaviour implied by the series, what is the distribution of adverse outcomes over a forward horizon? The engine does not forecast a price level or a direction. It simulates a large ensemble of forward paths and reports the *distribution* of three quantities of primary interest to an allocator: the expected maximum drawdown, the severity of losses in the extreme lower percentiles, and the time required to recover. Each reported statistic is accompanied by a confidence interval, and every run is exportable as a structured JSON object or a formatted PDF.

1.2 Limitations of stationary Monte Carlo

The standard approach draws returns from a single fixed distribution and simulates forward. It is analytically convenient but deficient in two respects that are material to risk measurement.

First, returns are not generated by a single stationary law. Volatility clusters, losses exhibit serial dependence, and the probability of a large decline conditional on a recent sequence of declines substantially exceeds its unconditional value. A model that ignores this structure understates sustained crisis drawdowns while overstating isolated single-day moves.

Second, tail observations are scarce. A fifteen-year equity history contains approximately four thousand daily observations, of which only a few hundred correspond to genuine stress. Any tail model estimated from those few hundred points carries substantial estimation error, and unadjusted simulation propagates that error without any accompanying measure of its reliability.

1.3 Contributions

The Blue Lotus engine addresses both deficiencies and incorporates an explicit model-risk diagnostic. This paper makes four contributions:

1. A fully specified, auditable risk engine (Sections 2–4) that structures simulation around a deterministic volatility-regime model, estimates tails with Extreme Value Theory, stabilises parameters with Bayesian shrinkage, and reports bootstrap confidence intervals on every metric.
2. The Model Fragility Index (Section 4.8), a Wasserstein-based, scale-invariant measure of how sensitive the output distribution is to parameter uncertainty.
3. A reproducible single-asset characterisation (Section 6) and, centrally, a strict out-of-sample event-study backtest across nine crises and ten assets, with a 213-cell calm-period calibration control (Section 7).

4. A transparent assessment (Sections 8–9) of the conditions under which the engine is well calibrated and those under which, by construction, no history-calibrated model can be accurate.

2. Engine Overview

The engine is a pipeline of seven stages. Each stage is a self-contained module with typed inputs and outputs, so that any stage may be inspected, replaced, or unit-tested in isolation.

1. **Input processing.** Clean the return series, winsorise extremes, and record raw moments for reporting.
2. **Regime detection.** Label every day calm, volatile, or crisis from rolling realised volatility and a negative-trend gate.
3. **Bayesian shrinkage.** Pull the noisy per-regime means and variances toward pooled estimates in proportion to how little data supports them.
4. **Tail estimation.** Fit a Generalized Pareto Distribution to the loss tail via Peaks-over-Threshold and derive analytical VaR and ES targets.
5. **Monte Carlo generation.** Simulate regime-switching paths, blended with drawdown-conditional dynamics and clamped by a loss floor.
6. **Measure change and metrics.** Optionally tilt the path measure to hit mean and ES targets (Esscher), then compute the drawdown, ES, and recovery distributions with bootstrap confidence intervals.
7. **Fragility analysis.** Re-run the generator under parameter perturbations and summarise the output’s sensitivity as a single index.

The result is serialised to JSON and rendered to a PDF tear-sheet, and the same core is exposed through a REST API and two front-ends (Section 5). All stages downstream of stage 1 are deterministic conditional on a fixed random seed, so any run is exactly reproducible — a property on which the backtesting protocol depends.

3. Design Rationale

Each modelling choice trades statistical sophistication against auditability. The engine consistently favours methods whose outputs admit a direct interpretation by a risk officer.

Deterministic regimes over a hidden Markov model. An earlier version of the engine employed a three-state Gaussian HMM estimated by Baum–Welch. Although theoretically attractive, it produced crisis fractions that varied substantially across assets and required manual

reseeding to avoid degenerate states — properties ill-suited to a production risk system. The current detector assigns regimes by explicit rules on rolling volatility and trend (Section 4.2). Each label admits a one-sentence justification, the crisis fraction is calibrated by construction through its percentile thresholds, and the model contains no latent state subject to misconvergence.

Extreme Value Theory for the tail. Parametric bulk distributions such as the Student- t constrain the extreme tail through the centre of the data. Peaks-over-Threshold EVT models only the exceedances, which is the asymptotically correct approach for extreme quantiles and the industry standard in quantitative risk management.

Bayesian shrinkage for stability. The crisis regime may hold only a few hundred days; its raw mean and variance are too noisy to drive a simulation. Conjugate shrinkage pulls each regime’s parameters toward pooled estimates by an amount that falls as the regime’s sample grows, trading a little bias for a large reduction in variance.

Esscher tilt as an optional corrector. When the simulated mean or Expected Shortfall departs from the EVT and regime targets, an Esscher exponential tilt reweights the paths to restore consistency. The tilt is optional and self-disabling: if its solver fails to converge, the engine reverts to identity weights and preserves the historical measure, rather than imposing an unstable correction to satisfy a target.

Bootstrap intervals throughout. A risk estimate reported without an interval conveys false precision. The engine resamples paths to attach a 90% confidence band to every reported scalar, so that estimation uncertainty is quantified explicitly rather than left implicit.

A fragility index for model risk. Even a well-calibrated model is only as trustworthy as it is stable. The Model Fragility Index quantifies how far the drawdown distribution moves under deliberate parameter perturbations, turning model risk into a reported number rather than an afterthought.

4. Methodology

4.1 Data preprocessing

Let $\{r_t\}_{t=1}^T$ be a daily return series. Prior to estimation, the series is validated as *decimal* returns rather than percentages. The admissible per-day range is set intentionally wide — up to an order-of-magnitude single-day move — so that legitimately high-volatility inputs, such as digital assets or individual securities subject to gap risk, are not rejected as malformed. Returns are then winsorised at the 1st and 99th percentiles, clipping extreme observations to those boundaries so that a small number of points cannot dominate the moment and tail estimates.

Raw mean, standard deviation, skewness, and excess kurtosis are recorded *prior* to winsorising for reporting. Annualised volatility is $\hat{\sigma}_{\text{ann}} = \hat{\sigma}_{\text{daily}}\sqrt{252}$.

4.2 Regime detection

Market conditions are classified into three states: calm (0), volatile (1), and crisis (2). For each $t > w$ the engine computes the w -day annualised realised volatility and the w -day rolling return,

$$\hat{\sigma}_t = \sqrt{252} \widehat{\text{std}}(r_{t-w}, \dots, r_{t-1}), \quad \tilde{r}_t = \sum_{j=t-w}^{t-1} r_j, \quad (1)$$

with $w = 20$ by default. Let q_{70}, q_{90} be the 70th and 90th percentiles of $\{\hat{\sigma}_t\}$. The raw label is 0 below q_{70} , 1 in $[q_{70}, q_{90})$, and 2 at or above q_{90} .

Two filters refine the crisis state. A *negative-trend gate* downgrades any crisis candidate with $\tilde{r}_t > 0$ to volatile, so high-volatility recovery rallies are not mislabelled as crises. A *persistence filter* downgrades any run of crisis days shorter than $d_{\min} = 5$ consecutive days, so isolated spikes do not count as regimes. The first w observations are set to calm.

From the final labels $\{s_t\}$ the transition matrix is estimated with Laplace (add-one) smoothing,

$$P_{ij} = \frac{1 + \sum_{t < T} \mathbf{1}[s_t = i, s_{t+1} = j]}{3 + \sum_{t < T} \mathbf{1}[s_t = i]}, \quad (2)$$

per-regime means and standard deviations are sample moments over each state, and the long-run distribution $\boldsymbol{\pi}$ is the dominant left eigenvector of $\mathbf{P}$, normalised to sum to one. A run whose crisis fraction exceeds 15% is flagged for review.

4.3 Bayesian shrinkage of regime parameters

The raw per-regime moments are stabilised by conjugate shrinkage. With a Gaussian–Gaussian model the posterior mean for regime k is

$$\mu_k^* = w_k \bar{r}_k + (1 - w_k) \mu_0, \quad w_k = \frac{n_k / \sigma^2}{n_k / \sigma^2 + 1 / \tau^2}, \quad (3)$$

where $\mu_0 = \bar{r}$ is the grand mean, σ^2 the pooled within-regime variance, τ^2 the between-regime variance, and n_k the regime's sample size. The weight w_k is the share of total precision coming from data, so sparse regimes shrink harder toward the grand mean. The variance uses an inverse-gamma update,

$$\sigma_k^{*2} = \frac{\beta_{\text{IG}} + \frac{1}{2} SS_k}{\alpha_{\text{IG}} + \frac{n_k}{2} - 1}, \quad SS_k = \sum_{t:s_t=k} (r_t - \bar{r}_k)^2, \quad (4)$$

with prior shape $\alpha_{\text{IG}} = 3$ and $\beta_{\text{IG}} = \sigma^2(\alpha_{\text{IG}} - 1)$, fixing the prior mode at the pooled variance.

4.4 Tail risk via Extreme Value Theory

Let $L_t = -r_t$ be losses. The Hill estimator of the tail index over the top k order statistics is

$$\hat{\xi}_k^{\text{Hill}} = \frac{1}{k} \sum_{i=1}^k [\log L_{(i)} - \log L_{(k+1)}], \quad (5)$$

and the threshold k^* is chosen where $\hat{\xi}_k$ is most stable across a sliding window, searching $k \in [0.02n, 0.15n]$. Given threshold $u^* = L_{(k^*+1)}$ and exceedances $Y_i = L_i - u^*$, the Generalized Pareto parameters $(\hat{\xi}, \hat{\beta})$ are fit by maximum likelihood to $F_u(y) = 1 - (1 + \xi y / \beta)^{-1/\xi}$, yielding analytical risk measures

$$\widehat{\text{VaR}}_\alpha = u^* + \frac{\hat{\beta}}{\hat{\xi}} \left[(\alpha n / N_u)^{-\hat{\xi}} - 1 \right], \quad \widehat{\text{ES}}_\alpha = \frac{\widehat{\text{VaR}}_\alpha + \hat{\beta} - \hat{\xi} u^*}{1 - \hat{\xi}}, \quad (6)$$

valid for $\hat{\xi} < 1$. If the fit fails or fewer than fifteen exceedances are available the engine falls back to empirical quantiles. Because the input has been winsorised at the 1st/99th percentile, the exceedances are bounded above, which on well-behaved indices drives the fitted shape negative (a short, bounded tail) — a point we return to in Section 6.

4.5 Monte Carlo path generation

Paths are simulated by a Markov regime switch: the next regime is drawn from row k_t of $\mathbf{P}$, and the return is drawn from the shrunken emission law $r_t^{(i)} \mid k_t = k \sim \mathcal{N}(\mu_k^*, \sigma_k^{*2})$. To inject path-dependence, each path's running drawdown state $d_t^{(i)} \in \{0, 1, 2\}$ (normal / moderate / severe, with thresholds -5% and -15%) blends the regime draw with a drawdown-conditional draw estimated from history:

$$r_t^{(i)} \leftarrow (1 - \omega_d) r_t^{(i)} + \omega_d \epsilon_t^{(d)}, \quad \epsilon_t^{(d)} \sim \mathcal{N}(\mu_d, \sigma_d^2), \quad (7)$$

with blend weights $\omega_0 = 0.10$, $\omega_1 = 0.30$, $\omega_2 = 0.50$, so paths in deeper drawdowns draw more heavily on the historical stress distribution. Finally a hard floor enforces simulation integrity: once a path's cumulative return falls below $f = -0.70$, all subsequent returns are set to zero,

$$r_t^{(i)} = 0 \quad \text{if} \quad C_{t-1}^{(i)} \leq f, \quad C_t^{(i)} = \sum_{s \leq t} r_s^{(i)}, \quad (8)$$

reflecting that a position down 70% would in practice be stopped out before compounding further.

4.6 Esscher change of measure

To align the simulated mean and tail with the regime and EVT targets, paths may be reweighted by an exponential (Esscher) tilt,

$$w_i \propto \exp(\lambda_1 \bar{r}_i + \lambda_2 \bar{r}_i \mathbf{1}[\bar{r}_i \leq \widehat{\text{VaR}}_\alpha]), \quad (9)$$

with (λ_1, λ_2) solving $\mathbb{E}_{\mathbb{Q}}[\bar{r}] = \mu^*$ and $\mathbb{E}_{\mathbb{Q}}[\bar{r} \mid \bar{r} \leq \widehat{\text{VaR}}_\alpha] = \text{ES}^*$. The system is solved from five random starts, taking the smallest-residual solution; if that residual exceeds 10^{-4} the tilt reverts to identity weights ($\lambda_1 = \lambda_2 = 0$), preserving the historical measure. Paths are then resampled with replacement in proportion to w_i .

4.7 Stress metrics and bootstrap intervals

From the path set the engine computes the maximum-drawdown distribution, the α -level Expected Shortfall (both aggregate and per-path), and the recovery time distribution. Every scalar θ carries a 90% interval from $B = 500$ path resamples,

$$\widehat{\text{CI}}_{90\%}(\theta) = [\hat{\theta}_{(0.05B)}^*, \hat{\theta}_{(0.95B)}^*]. \quad (10)$$

The bootstrap is fully vectorised so that 500 resamples cost roughly one metric evaluation.

4.8 Model Fragility Index

Definition 1 (Model Fragility Index). Let D_0 be the baseline maximum-drawdown distribution and D_θ the distribution under perturbation θ from a set Θ . Then

$$\text{MFI} = \frac{1}{|\Theta|} \sum_{\theta \in \Theta} \frac{W_1(D_0, D_\theta)}{\sigma(D_0)}, \quad W_1(\mu, \nu) = \int_0^1 |F_\mu^{-1}(u) - F_\nu^{-1}(u)| du, \quad (11)$$

where W_1 is the Wasserstein-1 distance and $\sigma(D_0)$ normalises by the baseline drawdown spread, making the index scale-invariant.

The perturbation set comprises five independent shocks — regime means $\times 1.10$ and $\times 0.90$, regime volatilities $\times 1.20$ and $\times 0.80$, and a 5% Dirichlet jitter of the transition rows — each run under the baseline seed so that only the parameter change, not sampling noise, moves the output. Grades follow $\text{MFI} < 0.25$ Robust, $0.25 \leq \text{MFI} < 0.65$ Moderate, $\text{MFI} \geq 0.65$ Fragile.

4.9 Multi-asset extension

For portfolios the engine estimates a Ledoit–Wolf shrinkage covariance, draws correlated regime innovations through a Cholesky factor, and aggregates path P&L at supplied weights, so joint

drawdowns reflect both marginal tails and cross-asset dependence. The validation in this paper is single-asset; the multi-asset path is left to future work.

5. System Architecture and Implementation

The engine is deployed as a production service rather than a research script alone. The numerical core (`engine/core.py`) is implemented in NumPy/SciPy and has no web dependencies. A FastAPI application (`api/`) wraps the core, submitting long-running computations as background jobs and persisting runs and results through an asynchronous SQLAlchemy layer (`db/`). The authentication subsystem — JWT tokens and API keys — is implemented in the codebase but disabled in the current build, which executes every request as a single shared workspace user; re-enabling per-account authentication requires a single dependency substitution. Results are serialised to a JSON-safe payload by a dedicated module (`engine/serializer.py`) and to a formatted tear-sheet by a ReportLab module (`reports/pdf.py`); both are exposed as direct exports in the interface. Two front-ends consume the API: a React single-page application (`web/`) for interactive use and a Streamlit application (`frontend/`) for expedited analysis. The identical codebase runs locally under Docker Compose and in production on Google Cloud Run, and every run is reproducible from its seed.

6. Single-Asset Characterisation: SPY

We first characterise the engine on the SPDR S&P 500 ETF (SPY), daily from January 2010 to June 2026, 4,138 observations, 17.1% annualised volatility, 10,000 paths over a 252-day horizon.

Regimes. The detector places 70.1% of days in calm, 24.0% in volatile, and 5.8% in crisis; the long-run stationary distribution is $(0.70, 0.24, 0.06)$. A crisis share near 6% is consistent with the observed frequency of sustained equity stress over the period (chiefly the March 2020 crash and the 2022 tightening cycle).

Tail. The Peaks-over-Threshold step used 275 exceedances beyond a daily loss threshold of 1.43%, with an implied 5% Expected-Shortfall target of -2.79% . As anticipated in Section 4, winsorising before the fit bounds the exceedances and yields a negative GPD shape ($\hat{\xi} = -1.26$, $\hat{\beta} = 0.020$): a short, bounded tail relative to the threshold rather than a Fréchet-type fat tail. For this sample the Esscher solver did not improve on the historical measure and reverted to identity weights, so the metrics below are reported under $\mathbb{P}$.

Risk metrics. Table 1 reports the core outputs with their 90% bootstrap intervals. The mean one-year maximum drawdown is -14.1% and the 5th-percentile drawdown is -37.0% ,

a reasonable tail benchmark for position sizing on a 17%-volatility index. The elevated never-recover rate requires qualification: it is a horizon artefact rather than a statement about the long-run index. Paths that enter drawdown late in the 252-day window have insufficient remaining horizon to recover and are recorded as unrecovered, although a longer horizon would in most cases capture the subsequent recovery.

Table 1: SPY risk metrics: 10,000 paths, 252-day horizon, seed 42.

Metric	Estimate	90% CI lower	90% CI upper
Mean max drawdown	−14.1%	−14.2%	−13.9%
Median max drawdown	−11.0%	—	—
5th-percentile drawdown	−37.0%	−37.8%	−36.3%
Aggregate ES (5%)	−1.80%	−1.81%	−1.80%
Median recovery time	35 days	—	—
Never recover (252d)	53.2%	—	—
Model Fragility Index	0.25	Moderate	

Fragility. The MFI of 0.25 sits on the Robust/Moderate boundary. Volatility perturbations dominate the index (W_1 of 0.043 and 0.041 for $\pm$ shocks, versus 0.005–0.008 for mean shocks and 0.033 for transition jitter), as expected since the drawdown distribution is driven primarily by the volatility of the return process.

7. Out-of-Sample Backtest and Verification

The single-asset section establishes that the engine produces internally coherent estimates. Internal coherence, however, is a necessary but not a sufficient condition. The material question is whether the predicted distributions in fact contained the realized outcomes. This question is addressed strictly out-of-sample.

7.1 Protocol

For an asset with returns $r_{1:T}$ and dates $d_{1:T}$ and a crisis window $[t_0, t_1]$, the *training set* is $\{r_i : d_i < t_0\}$ — strictly pre-crisis — and we require at least 504 training observations. The engine is fit on the training set (sensitivity and its internal validator disabled, seed 42, $N = 5000$ paths) with horizon H equal to the number of trading days in $[t_0, t_1]$, so the predicted maximum-drawdown distribution and the realized maximum drawdown span the same length. All drawdowns are computed in additive cumulative-return space, the engine’s internal convention, applied identically to simulated and realized paths.¹

¹The additive maximum drawdown of a path is $\min_t (C_t - \max_{s \leq t} C_s)$ with $C_t = \sum_{i \leq t} r_i$. Because it does not compound it can fall below -100% in extreme cases; in price terms the figures are smaller (e.g. financials in 2008 troughed near -83%).

7.2 Coverage statistics

For each cell we record the realized maximum drawdown DD^{real} and the simulated drawdown distribution $\{DD^{(j)}\}_{j=1}^N$, and compute the *probability-integral-transform rank*

$$u = \frac{1}{N} \sum_{j=1}^N \mathbf{1}[DD^{(j)} \leq DD^{\text{real}}], \quad (12)$$

the fraction of simulated paths at least as bad as reality. Under a correct model $u \sim \text{Uniform}(0, 1)$. A $p5$ ($p1$) breach occurs when DD^{real} is worse than the simulated 5th (1st) percentile. For x breaches in n cells at level p the Kupiec proportion-of-failures statistic

$$\text{LR}_{\text{POF}} = -2 \ln \frac{(1-p)^{n-x} p^x}{(1-\hat{\pi})^{n-x} \hat{\pi}^x}, \quad \hat{\pi} = x/n, \quad (13)$$

is asymptotically χ_1^2 . The calm control provides the regime in which the test's nominal-rate assumption is appropriate; on the deliberately adversarial crisis set the statistic quantifies the conditional exceedance rate rather than functioning as an acceptance test.

7.3 Data universe

Ten liquid proxies span equity beta (SPY, QQQ, IWM, DIA, EFA), sector concentration (XLF, XLE), duration (TLT), credit (HYG), and a diversifier (GLD), each pulled to the report date through the engine's own loader. Nine crisis windows run from the 2008 financial crisis through the 2023 regional-bank episode. The grid yields 89 crisis cells (one skipped: high-yield credit in 2008, which lacks the required pre-crisis history) and 213 calm-control cells from 22 non-overlapping six-month windows.

7.4 Calm-period calibration

Table 2 and Figure 1 establish the control. On calm windows the engine breaches its 5% bound in 2.8% of cells and its 1% bound in 0.9%, both statistically indistinguishable from nominal (Kupiec $p = 0.11$ and 0.93). The mean PIT rank of 0.69 indicates that the engine is mildly conservative on ordinary data, predicting marginally more risk than is realized. For a risk system this is the prudent direction of error, and it is a precondition for interpreting the crisis results: a model that generated excessive exceptions under calm conditions would provide no basis for assessing its behaviour under stress.

7.5 Crisis event study

The crisis results (Table 3, Figures 2–3) invert the control. Coverage degrades monotonically with severity: the 2015–16 China/oil selloff is fully covered and the 2023 regional-bank episode

Table 2: Headline coverage: calm control versus crisis event study.

	Calm control	Crisis study	<i>Calibrated target</i>
Cells evaluated	213	89	—
$p5$ breach rate	2.8%	56.2%	$\approx 5\%$
Kupiec p -value	0.11	$< 10^{-40}$	> 0.05
$p1$ breach rate	0.9%	31.5%	$\approx 1\%$
Kupiec p -value	0.93	$< 10^{-33}$	> 0.05
90% band coverage	71.8%	42.7%	$\approx 90\%$
Mean PIT rank	0.69	0.14	0.50

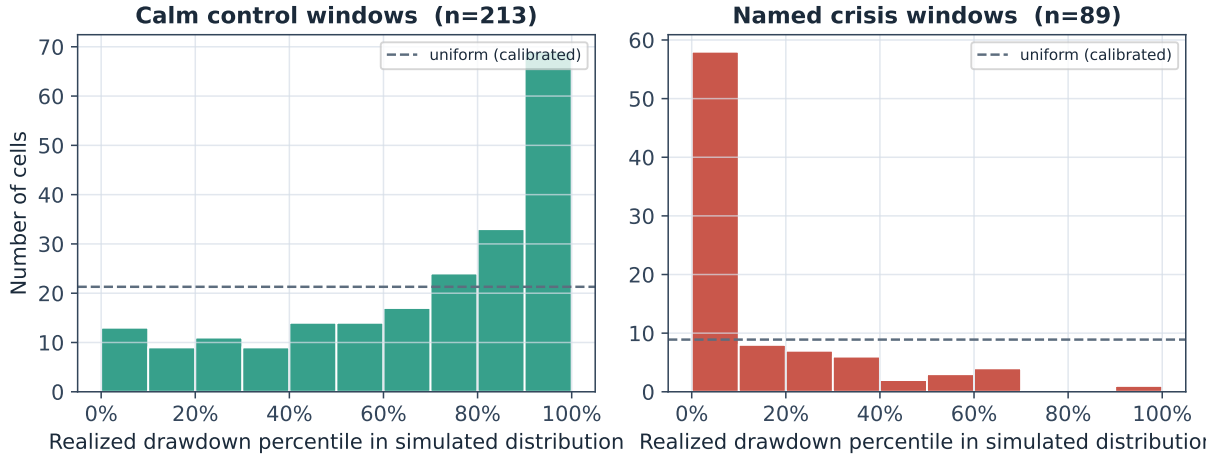


Figure 1: Realized-drawdown PIT ranks. Calm windows (left) track the uniform benchmark; crisis windows (right) pile at the left edge, where realized losses fall in the engine’s lower tail.

substantially so, whereas every COVID-2020 cell and nearly every 2008 cell breaches even the 1% bound. Realized crisis drawdowns exhibit a mean PIT rank of 0.14, deep within the predicted lower tail, and the two most severe systemic shocks fall essentially at rank zero, the most extreme exceedance the statistic can record.

7.6 Cross-asset behaviour

Sorting by mean PIT rank (Table 4) shows the gap concentrates in high-beta and cyclical exposures — energy, small caps, developed-international, broad equity — which suffer the most violent crisis drawdowns. Defensive and diversifying sleeves behave very differently: long-duration Treasuries (rank 0.30) and gold (rank 0.35) sit near the centre of the predicted distribution, and high-yield credit is breached in only two of eight cells. The miss is a faithful reflection of which exposures carry systemic tail risk, not uniform model error.

Table 3: Per-crisis coverage. “Mean DD” and “Pred. $p5$ ” are cross-asset averages (additive); “Rank” is the mean realized PIT rank (lower = deeper miss).

Crisis	n	Mean DD	Pred. $p5$	$p5$ br.	$p1$ br.	Cov.	Rank
2010 Flash Crash / Euro I	10	−14.3	−17.2	4/10	0/10	60%	0.20
2011 US Downgrade / Euro II	10	−19.4	−16.7	7/10	2/10	30%	0.09
2015–16 China / Oil Selloff	10	−16.8	−24.6	0/10	0/10	100%	0.27
Feb 2018 Volmageddon	10	−8.5	−6.7	8/10	3/10	20%	0.06
Q4 2018 Selloff	10	−18.5	−17.0	6/10	4/10	30%	0.19
March 2020 COVID Crash	10	−37.7	−9.9	10/10	10/10	0%	0.00
2022 Rate-Shock Bear	10	−27.7	−29.6	4/10	1/10	60%	0.09
2023 Regional Bank Crisis	10	−4.8	−7.4	3/10	0/10	70%	0.32
2008 Global Financial Crisis	9	−52.2	−26.4	8/9	8/9	11%	0.01

Table 4: Per-asset coverage across crises, deepest to shallowest tail miss.

Asset	n	Mean DD	Pred. $p5$	$p5$ br.	Cov.	Rank
Energy (XLE)	9	−32.9	−20.9	7/9	22%	0.03
Russell 2000 (IWM)	9	−29.4	−20.0	7/9	22%	0.04
Developed ex-US (EFA)	9	−24.2	−17.0	7/9	22%	0.06
S&P 500 (SPY)	9	−22.2	−14.2	7/9	22%	0.07
Financials (XLF)	9	−33.2	−24.2	6/9	33%	0.07
Dow 30 (DIA)	9	−20.7	−15.0	5/9	44%	0.10
High-Yield Credit (HYG)	8	−9.9	−9.6	2/8	75%	0.16
Nasdaq 100 (QQQ)	9	−23.3	−24.8	3/9	67%	0.19
20Y+ Treasuries (TLT)	9	−10.4	−10.1	4/9	56%	0.30
Gold (GLD)	9	−11.2	−15.0	2/9	67%	0.35

7.7 Exemplar cells

Four cells illustrate the range of outcomes. **SPY in COVID-2020**: realized −38.2% against a predicted $p5$ of −8.6% ($p1$ −11.9%), rank 0.000; the most rapid 30% equity drawdown on record lies outside the support implied by pre-2020 data. **XLF in 2008**: realized −103.6% (additive) against $p5$ −38.2%, rank 0.000, reflecting a solvency crisis internal to the financial sector. **SPY in 2015–16**: realized −13.3% against $p5$ −20.3%, rank 0.25, a correction fully contained within the predicted distribution. **SPY in 2023**: realized −3.4% against $p5$ −6.7%, rank 0.28, again contained and correctly bounded by the engine.

8. Discussion

The two regimes support a single conclusion. The calm control demonstrates that the engine is correctly specified for ordinary conditions: at both the 5% and the 1% tail level its out-of-sample exception rates are statistically indistinguishable from nominal, with error biased toward

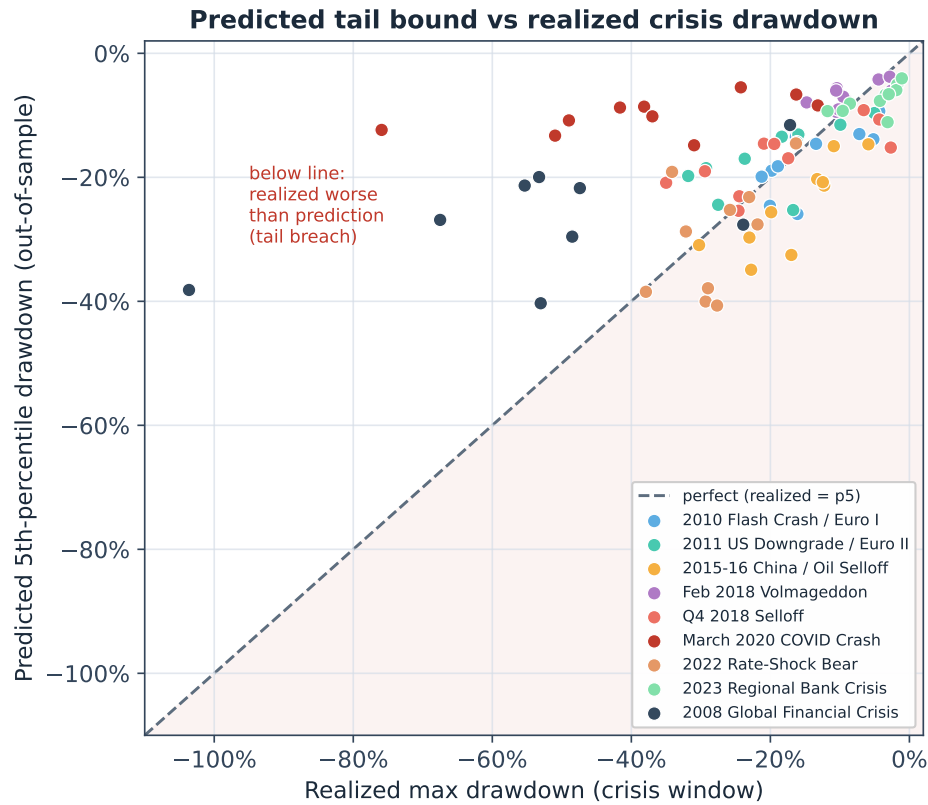


Figure 2: Predicted 5th-percentile drawdown versus realized crisis drawdown, one point per cell. Points below the identity line are tail breaches; the lower-left cluster is COVID-2020 and 2008.

conservatism. The crisis study demonstrates that the outcomes which do breach are precisely those that a history-calibrated model cannot anticipate — the most abrupt and reflexive systemic shocks, in which volatility, correlation, and liquidity dislocate more rapidly than any trailing estimation window can represent. Prior to March 2020, the March 2020 event was absent from the data.

This is not a defect specific to Blue Lotus; it is a property of the estimation problem. Any model whose tail is calibrated to realized returns is bounded by its worst observed history. The contribution of the backtest is to *quantify* the residual gap rather than to obscure it, and to establish that the gap is monotone in severity and concentrated in the exposures that carry systemic tail risk. Accordingly, the production engine does not rely on the empirical tail alone: its crisis-regime structure, drawdown-conditional blend, and operator-configurable loss floor are designed to extend the simulated distribution into the region that the backtest identifies as beyond historical reach. The appropriate deployment is therefore to apply the engine as calibrated here for ordinary risk measurement and to engage the crisis overlay when sizing for systemic scenarios.

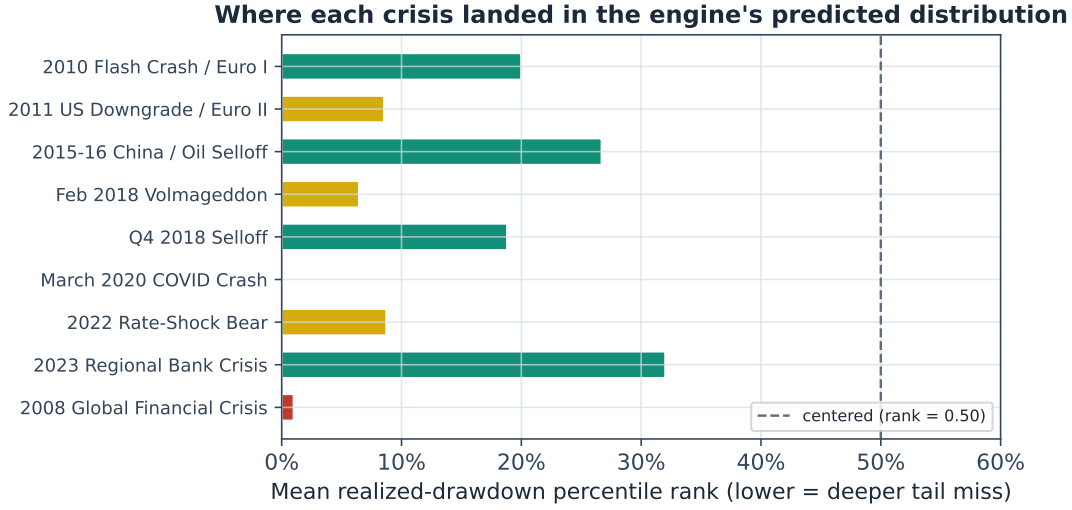


Figure 3: Mean realized-drawdown percentile rank by crisis. Bars near 0.50 mean the engine centred the outcome; bars near zero mean it fell in the extreme tail.

9. Limitations

1. **Adversarial crisis sample.** The nine windows are chosen because they are extreme, so the crisis breach rates estimate a conditional, not an unconditional, exception rate; the unconditional estimate is the calm control.
2. **Additive drawdown convention.** Drawdowns are summed, not compounded, to match the generator; this inflates very deep drawdowns relative to price terms but is applied identically on both sides of every comparison.
3. **Horizon matching.** Setting H to the realized window length is the cleanest like-for-like choice but evaluates short, sharp episodes over short horizons where the simulated distribution is itself narrow.
4. **Numerical fallbacks.** The Esscher tilt and the GPD fit both fall back to conservative defaults (identity weights, empirical quantiles) when their solvers do not converge; these events are logged.
5. **Single data vendor.** All prices come from one adjusted-close source; vendor differences in corporate-action handling would shift individual cells modestly but not the aggregate picture.
6. **Univariate validation.** The backtest is single-asset; the multi-asset dependence machinery is described but not validated here.

10. Conclusion and Future Work

The Blue Lotus engine maps a return series to a fully characterised, bootstrap- bounded distribution of drawdown, tail-loss, and recovery outcomes, constructed from auditable components and delivered as a reproducible production service. Under a strict out-of-sample protocol it is statistically well calibrated under ordinary market conditions — not rejectable at either the 5% or the 1% tail level across 213 calm cells — and its behaviour under the nine largest crises of the past two decades is fully characterised, with coverage that degrades monotonically in severity and a discrete failure only at the two most severe systemic shocks. This is the appropriate standard for such a system: not that it predict unprecedented events, but that it delineate precisely the boundary of its validity.

Three extensions follow directly. The regime detector could incorporate exogenous stress signals — implied volatility, credit spreads, and funding indicators — to improve crisis identification in advance of realized volatility. The multi-asset dependence model warrants the out-of-sample treatment applied here to the univariate case, validated against joint drawdown events. Finally, the Model Fragility Index could itself be backtested, by testing whether high-fragility periods precede realized instability. In each case the evaluative standard is the one adopted throughout: quantify the residual gap, and report it.

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A. Default Parameters

Table 5: Engine default parameters used throughout this paper.

Module	Parameter	Default
Preprocessing	accepted daily range	up to ± 10.0 (decimal)
Preprocessing	winsorisation	1st / 99th percentile
Regime	rolling window w	20 days
Regime	vol thresholds	70th / 90th percentile
Regime	min crisis duration	5 days
Shrinkage	IG prior shape	$\alpha_{IG} = 3$
Tail	EVT level α	0.05
Tail	Hill search range	$[0.02n, 0.15n]$
Monte Carlo	paths N	10,000 (5,000 in backtest)
Monte Carlo	horizon	252 days (window length in backtest)
Monte Carlo	drawdown blend	0.10/0.30/0.50
Monte Carlo	loss floor f	-0.70
Esscher	residual tolerance	10^{-4}
Bootstrap	resamples B	500
MFI	grades	0.25 / 0.65
All	random seed	42

B. Reproducibility

The single-asset characterisation, the crisis event study, the calm control, the aggregate tables, and the figures all regenerate from `backtest/oos_backtest.py`, `backtest/control_backtest.py`, `backtest/aggregate.py`, and `backtest/make_figures.py` with fixed seed 42. Raw per-cell output is written to `backtest/results/` as JSON.